

**Home bias and international risk sharing:
Twin puzzles separated at birth**

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Introduction

Hedging is a central topic in macroeconomics and finance, but macroeconomics and financial literature tend to have different notions of full hedging. The macroeconomics literature departs from the benchmark model of perfect markets, which under standard assumption implies that consumption growth rates are equalized ("perfect risk sharing"). While the financial literature typically departs from the benchmark model Capital Asset Pricing Model (CAPM), which under standard assumption predicts that countries hold identical international portfolios of risky assets. The article measures the deviation from the perfect risk sharing allocation (or, equivalently, the amount of risk sharing obtained), also the deviation from the international CAPM allocation ("home bias").

Home bias and risk sharing may be manifestations of the same underlying behavior: if agents diversify their portfolios internationally, they will likely obtain smoother income streams as domestic shocks partially will be offset by foreign asset income and smoother income is likely to imply smoother consumption. However, Home bias and risk sharing need not be close twins. As explained by Lewis (1999), there are several circumstances where home bias may not lead to lack of risk sharing. In particular, even if agents do not smooth income through cross-ownership of assets they can smooth consumption through borrowing and lending. Also, full international diversification of equity portfolios may not lead to smooth income if overall equity investment is small relative to GDP.

Recent research in finance builds on the mean-variance framework, which implies that investors prefer portfolios with high mean returns and low variance. In an ideal scenario without market frictions and with the presence of a safe asset, the simplest CAPM model predicts that all investors will hold a combination of the safe asset and market portfolio (risky assets). Even without a safe asset, if investors have similar mean-variance utility trade-offs, they will hold the same world portfolio.

Home bias may occur due to currency risk, Transaction costs associated with international and a third explanation centers on information asymmetry (Lack of information increases the risk of foreign investment). The strong decline in equity and debt home bias during the late 1990s is consistent with a role for declining costs of trading goods and acquiring information. While currency risk has been eliminated by mutual investments among members of the European Monetary Union, countries that are not members of any currency union also display rapidly declining home bias as Norway.

The paper empirically provides the missing link between the home bias and risk sharing by demonstrating that disappearing home bias and increasing risk sharing indeed move hand-in-hand by using a panel of

countries from the (OECD). Through using two alternative measures of risk sharing the first is based on macroeconomic literature that focuses on consumption risk sharing and the second measure is the income risk sharing based on the GNI.

International portfolio holdings

Table 1, display holdings of foreign equity, debt, and foreign direct investment for 1993 and 2003 for the 24 OECD countries that comprise the sample: Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Greece, Iceland, Ireland, Italy, Japan, Mexico, Netherlands, New Zealand, Norway, Portugal, Spain, Sweden, Switzerland, Turkey, United Kingdom, and the United States. The foreign assets and liability holdings of each country are normalized by its nominal GDP. It is immediately clear that holdings of foreign assets have increased steeply from 1993 to 2022. Table 1 also reveals large differences across countries. For example, in 2022 Ireland held large amounts of foreign equity and debt with levels that far exceeded the level of their GDP. Ireland is somewhat of an outlier with larger holdings of foreign stocks and bonds relative to GDP than other countries; the next highest ratio of foreign equity to GDP is found for Switzerland. At the other end of the spectrum, Turkey held foreign equity and debt in amounts less than 1 percent of Turkey's GDP while Japan held an amount of foreign equity equal to only 6 percent of GDP in 2003 and about 43 percent in 2022.

Foreign direct investment assets are on average larger than portfolio equity holdings. Liability holdings are quite like asset holdings although debtor nations, such as Australia, holds less assets than liabilities and vice versa for creditor nations, such as Switzerland. The striking aspect of the numbers in Table 1 is the steep increase in foreign asset holdings from 1993 to 2022.

Home bias

Based on the international version of the CAPM, Home Bias is 0 if the share of country's equity investment that is invested domestically equals the share of the country's foreign equity market in other words, a country will have Home Bias equal to 0 if it shows no preference for equity issued domestically. Equity Home Bias is 1 if the country invests 100 percent domestically.

In Table 2, the left-most columns show the percentage share of foreign equity in the aggregate portfolio of each country. Foreign equity holdings have increased faster than the overall domestically held portfolios. Overall, the average Equity Home Bias has decreased from 0.84 in 1993 to 0.55 in 2022. In 2022 several

countries had Equity Home Bias less than 0.5, namely the United Kingdom, Germany and Austria. Equity Home Bias in these countries has declined rapidly, for example, in Austria Equity Home Bias has declined steeply from 0.87 to 0.23 from 1993 to 2022. For Debt Home Bias, the numbers are fairly like those for Equity Home Bias. For example, the average Debt Home Bias is 0.64 in 1993 and 0.52 in 2003. Debt Home Bias has declined for most countries, with the exceptions being Japan, Mexico, and Turkey. All countries have positive Debt Home Bias, but Ireland has the lowest at only 0.10 in 2003.

Table 1	Portfolio equity assets ratio			Debt assets ratio			FDI assets ratio			Portfolio equity liabilities ratio			Debt liabilities ratio			FDI liabilities ratio		
Year	1993	2003	2022	1993	2003	2022	1993	2003	2022	1993	2003	2022	1993	2003	2022	1993	2003	2022
Australia	0.0696	0.1718	0.4262	0.0845	0.1847	0.3699	0.1469	0.3400	0.4120	0.1162	0.2569	0.3319	0.4758	0.6954	0.7938	0.2834	0.3954	0.4838
Austria	0.0196	0.1684	0.3844	0.4648	1.2481	1.0879	0.0426	0.2223	0.6961	0.0189	0.1010	0.1433	0.6243	1.5268	1.3340	0.0599	0.2156	0.5796
Belgium	0.2712	0.4611	0.8203	1.4842	2.4156	1.5274	0.2864	0.9872	1.8460	0.0377	0.0877	0.3444	1.5353	2.3014	1.7678	0.4311	1.1231	1.5675
Canada	0.1203	0.3081	0.7356	0.1729	0.2163	0.7575	0.2618	0.5890	1.0244	0.0708	0.1924	0.3287	0.6682	0.6168	1.2054	0.3251	0.5194	0.7267
Denmark	0.0557	0.2390	0.9890	0.5383	0.8103	1.0091	0.1371	0.4704	0.9116	0.0227	0.1433	0.9311	0.9607	1.2384	0.9974	0.1289	0.4594	0.5995
Finland	0.0035	0.2103	0.9920	0.2855	0.8020	1.3703	0.1445	0.5427	0.7140	0.0588	0.6020	0.6862	0.8707	0.9095	1.9077	0.0889	0.3924	0.5158
France	0.0392	0.1845	0.3333	0.4996	1.0446	1.8452	0.2060	0.4068	0.7355	0.0784	0.2662	0.4021	0.5680	1.1439	2.2810	0.2320	0.2898	0.5222
Germany	0.0556	0.2304	0.4402	0.4554	1.0794	1.4878	0.0756	0.3518	0.7419	0.0452	0.1449	0.2676	0.4423	1.2169	1.2834	0.0343	0.3301	0.4942
Greece	0.0036	0.0217	0.0568	0.1750	0.4542	1.2710	0.0069	0.0640	0.0992	0.0133	0.0798	0.1150	0.3757	1.0107	2.6233	0.0857	0.1140	0.2537
Iceland	0.0035	0.2785	0.6194	0.0353	0.3631	0.2210	0.0102	0.1787	0.2225	0.0001	0.0256	0.0737	0.6468	1.4036	0.5855	0.0106	0.1314	0.3741
Ireland	0.2594	1.3515	3.5848	0.7924	6.4211	8.0367	0.5609	1.1670	3.3976	0.3177	2.9260	8.0205	1.0744	4.2241	4.5273	0.6590	2.0752	3.7125
Italy	0.0285	0.2140	0.5535	0.2926	0.5928	0.6583	0.0782	0.1628	0.3728	0.0379	0.1356	0.1335	0.4493	0.8650	1.2056	0.0521	0.1538	0.3233
Japan	0.0239	0.0607	0.4339	0.3769	0.5115	1.0330	0.0572	0.0757	0.4888	0.0377	0.1241	0.4027	0.3041	0.2940	1.0136	0.0037	0.0213	0.0821
Mexico	0.0034	0.0123	0.0435	0.0481	0.0829	0.1297	0.0161	0.0630	0.1636	0.0916	0.0738	0.1004	0.2497	0.2205	0.3102	0.0821	0.2063	0.4860
Netherlands	0.1676	0.6128	1.0158	0.7710	1.9492	2.4280	0.3375	3.7498	6.1622	0.2595	0.5611	1.0276	0.7573	2.8707	2.6095	0.2092	2.9615	5.2240
New Zealand	0.0141	0.2169	0.3852	0.0466	0.2407	0.2717	0.2210	0.2482	0.1048	0.0831	0.1088	0.1735	0.6558	0.8522	0.7341	0.4690	0.5672	0.4186
Norway	0.0422	0.3329	1.7906	0.2129	0.9558	1.4505	0.1046	0.3602	0.4765	0.0793	0.0981	0.2414	0.3673	0.9686	1.1873	0.1130	0.2135	0.3885
Portugal	0.0113	0.0754	0.2021	0.2662	1.2736	0.9131	0.0225	0.3028	0.3622	0.0308	0.2266	0.1704	0.3702	1.7014	1.4822	0.1612	0.4769	0.8229
Spain	0.0112	0.0893	0.3018	0.2722	0.7270	0.9604	0.0482	0.2677	0.5863	0.0431	0.2059	0.2104	0.3838	0.9870	1.5494	0.1545	0.4863	0.7628
Sweden	0.0707	0.4264	1.0794	0.2551	0.5382	0.7883	0.2092	0.7668	1.1234	0.1024	0.2684	0.5361	0.8521	1.0544	1.2928	0.0605	0.6687	0.9270
Switzerland	0.3269	0.8108	1.1099	1.5651	2.8521	1.9880	0.3578	1.0658	2.5381	0.5112	1.1116	1.5966	0.8854	1.9397	2.0000	0.2032	0.6317	2.1866
Turkey	0.0026	0.0045	0.0036	0.0831	0.1211	0.1352	0.0010	0.0195	0.0618	0.0097	0.0284	0.0318	0.2720	0.4467	0.4327	0.0359	0.1056	0.2258
United Kingdom	0.2485	0.3659	0.7404	1.2295	2.1136	2.6097	0.2210	0.7560	0.8364	0.1712	0.5328	0.6490	1.3539	2.3067	2.6438	0.1740	0.4292	1.0148
United States	0.0793	0.1815	0.4047	0.1927	0.2899	0.3455	0.1756	0.2651	0.3643	0.0545	0.1606	0.4734	0.3020	0.5509	0.8253	0.1329	0.2412	0.4816
Average	0.08	0.29	0.73	0.44	1.14	1.36	0.16	0.56	1.02	0.10	0.35	0.72	0.64	1.31	1.52	0.17	0.55	0.97

Table 2	Foreign Equity in Portfolio %			Equity Home bias			Foreign debt in Portfolio %		Debt Home bias	
	1993	2003	2022	1993	2003	2022	1993	2003	1993	2003
Australia	11.33	17.23	39.44	0.89	0.83	0.61	15.32	21.71	0.85	0.78
Austria	14.12	59.50	76.52	0.86	0.41	0.23	47.24	61.68	0.53	0.38
Belgium	45.97	50.17	--	0.54	0.50	--	51.11	66.94	0.49	0.33
Canada	11.16	27.21	43.51	0.89	0.73	0.56	15.66	19.25	0.84	0.81
Denmark	17.85	36.57	--	0.82	0.63	--	25.19	29.27	0.75	0.71
Finland	1.66	-74.44	--	0.98	--	--	29.32	66.83	0.71	0.33
France	12.86	28.23	--	0.87	0.72	--	39.23	51.97	0.61	0.48
Germany	24.87	44.57	69.31	0.75	0.55	0.31	40.57	56.05	0.59	0.44
Greece	--	4.62	26.92	--	0.95	0.73	23.87	41.02	0.76	0.59
Iceland	--	--	--	--	--	--	8.41	20.90	0.92	0.79
Ireland	--	--	--	--	--	--	63.49	90.02	0.37	0.10
Italy	22.07	45.70	--	0.78	0.54	--	20.50	32.05	0.79	0.68
Japan	3.82	10.29	33.35	0.96	0.90	0.67	31.26	22.82	0.69	0.77
Mexico	1.16	12.48	17.19	0.99	0.88	0.83	34.46	24.49	0.66	0.76
Netherlands	40.03	68.53	--	0.60	0.31	---	54.86	65.74	0.45	0.34
New Zealand	2.80	42.60	62.66	0.97	0.57	0.37	11.83	45.64	0.88	0.54
Norway	22.23	50.99	--	0.78	0.49	---	34.62	69.12	0.65	0.31
Portugal	--	37.38	--	--	0.63	--	37.43	67.40	0.63	0.33
Spain	5.85	13.05	53.85	0.94	0.87	0.46	36.69	53.28	0.63	0.47
Sweden	16.05	41.60	--	0.84	0.58	--	21.76	39.29	0.78	0.61
Switzerland	40.28	47.51	63.43	0.60	0.52	0.37	71.43	82.24	0.29	0.18
Turkey	1.83	2.33	1.06	0.98	0.98	0.99	49.12	21.36	0.51	0.79
United Kingdom	23.18	36.17	67.56	0.77	0.64	0.32	77.19	84.03	0.23	0.16
United States	10.03	14.33	26.73	0.90	0.86	0.73	12.84	16.58	0.87	0.83
Average	16.46	32.91	44.73	0.84	0.67	0.55	35.56	47.90	0.64	0.52

- Equity Home Bias = $[1 - \text{foreign equity in portfolio}]$. Foreign equity in portfolio = total foreign equity held by the country / country total equity portfolio, where the total equity in portfolio = stock market capitalization + foreign equity held by the country – equity held by foreigners.
- Debt Home Bias = $[1 - \text{foreign debt in portfolio}]$. Foreign debt in portfolio = total foreign debt held by the country / country total debt portfolio, where the total debt in portfolio = domestic debt security outstanding + total foreign debt security assets held.

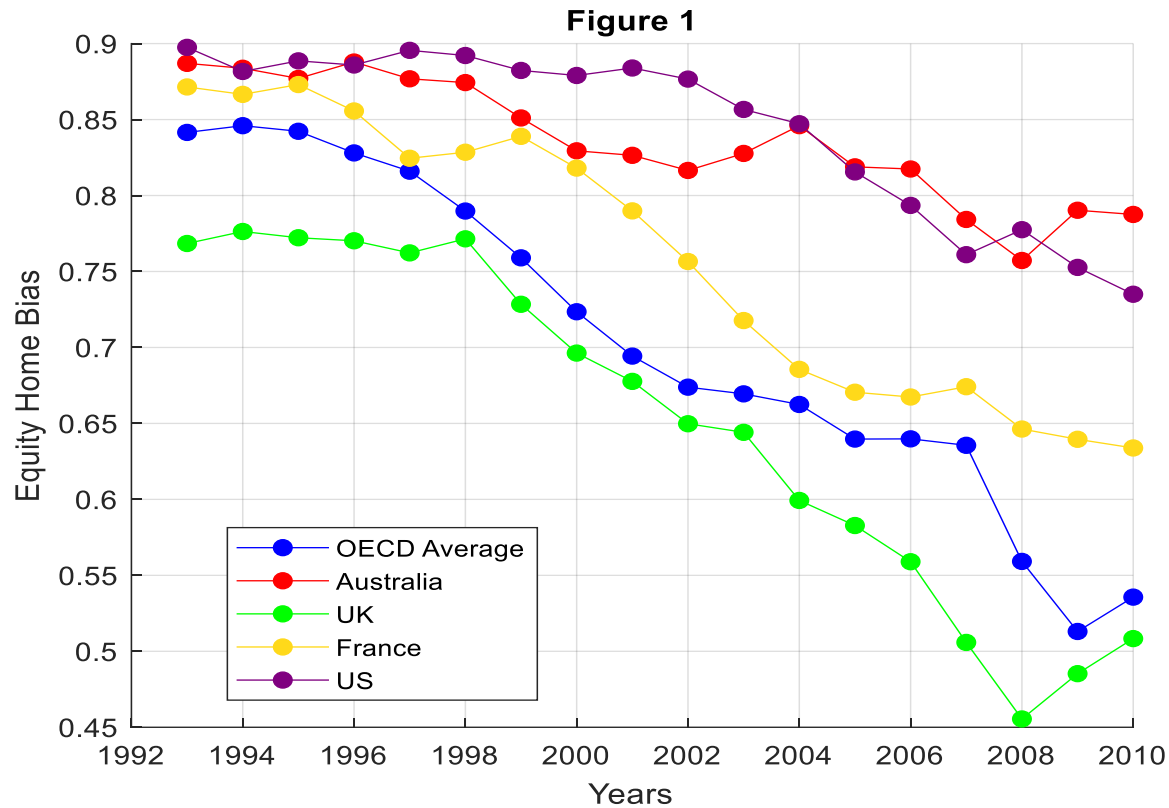


Figure 1 displays the average level of equity home bias for each year of the sample and the equity home bias for Australia, UK, France, and the US. The graph shows that home bias on average is steadily declining with an increase after the 2008 financial crisis. The next section determines whether this has been associated with increasing international risk sharing.

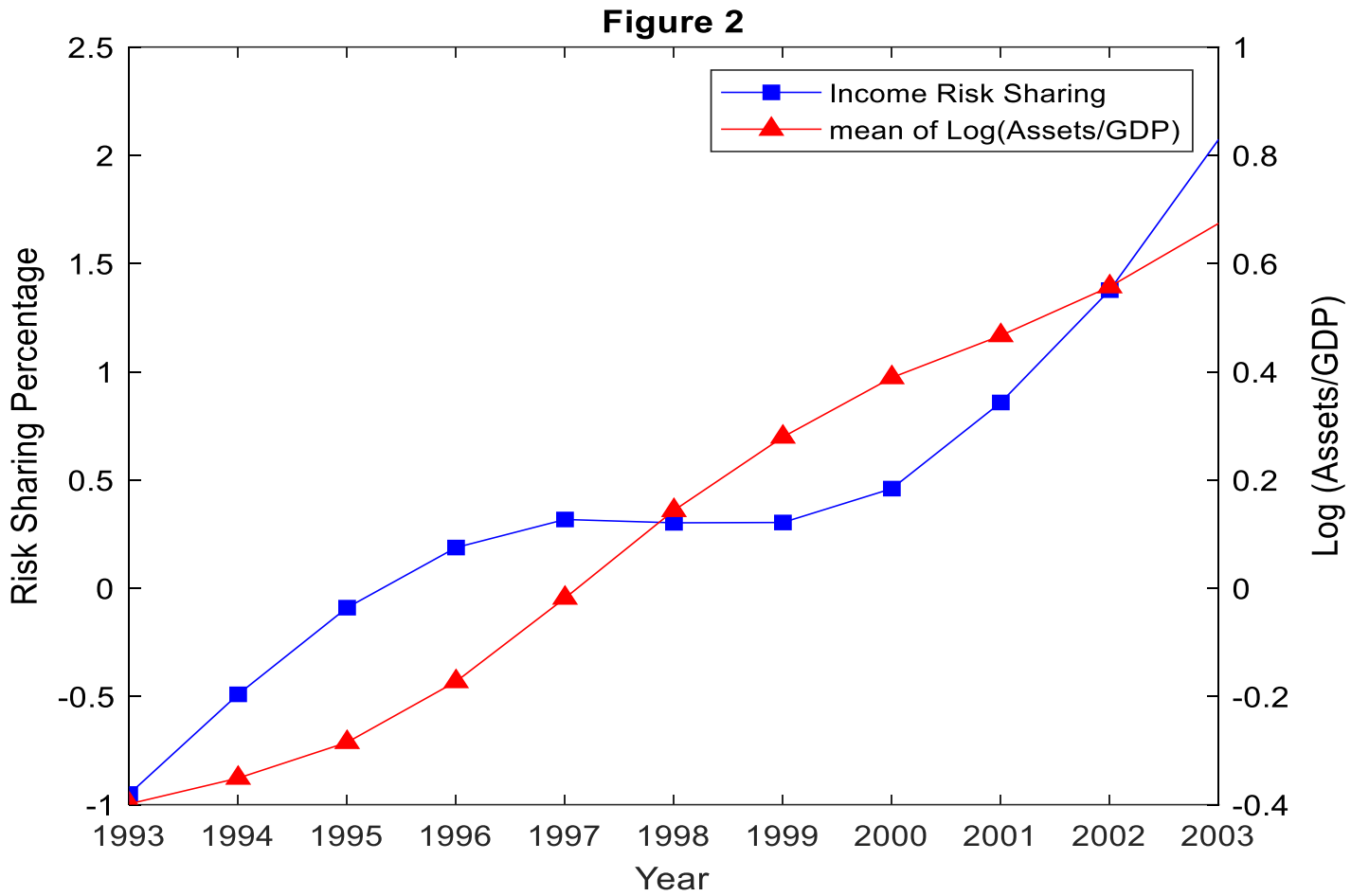
International Risk Sharing

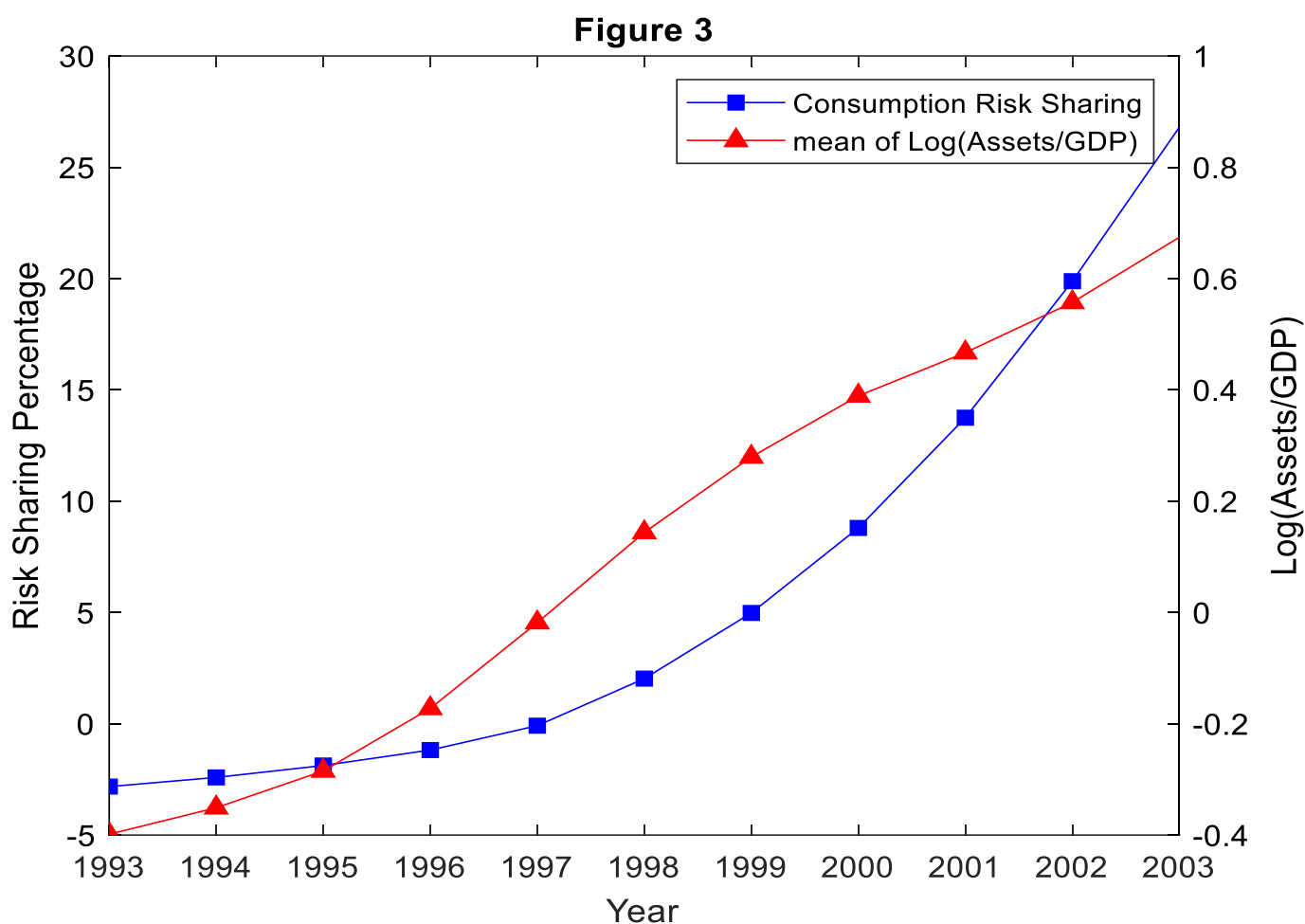
The macroeconomic literature focuses on aggregate consumption and income patterns, where consumption growth rates in all countries are identical as “full (or perfect) consumption risk sharing (or consumption smoothing).” This condition will hold for endowment economies if consumers have identical Constant Relative Risk Aversion utility functions and access to a complete set of Arrow-Debreu markets. Similarly, the term “full (or perfect) income risk sharing (or income smoothing)” describes the situation where the growth rate of GNP is identical in all countries, in this case, we would expect consumption growth rates to also be similar. Risk sharing year-by-year deviations from perfect income and consumption smoothing are quantified through cross-sectional regressions:

$$\Delta \log (GNI_{it}) - \Delta \log (GNI_t) = \text{constant} + \beta_{K,t} (\Delta \log (GDP_{it}) - \Delta \log (GDP_t)) + \epsilon_{it}$$

$$\Delta \log(\text{CONS}_{it}) - \Delta \log(\text{CONS}_t) = \text{constant} + \beta_{c,t} (\Delta \log(\text{GDP}_{it}) - \Delta \log(\text{GDP}_t)) + \epsilon_{it}$$

where CONS_{it} is country i 's year t per capita consumption, GNI_{it} is the year t per capita aggregate gross national income for the group. In this regression, subtracting from each variable the aggregate value is crucial because aggregate GDP-growth of the group is not insurable. The regression coefficient measures the average co-movement of countries' idiosyncratic GNP and consumption growth with GDP growth, a lower β indicates greater income and consumption smoothing. Under perfect risk sharing, the left-hand side of equation (2) will be zero which implies β will be zero.





Mean of $\log(\text{assets}/\text{GDP})$ is the cross sectional mean of foreign (equity + debt + FDI) holdings normalized by GDP for 24 OECD countries. Graphical evidence, such as Figure 2, displays year-to-year income smoothing measures and assets holdings normalized by GDP for OECD countries, revealing increasing income risk sharing through the 1990s. This trend correlates with rising international asset holdings. Figure 3 shows similar patterns for consumption smoothing, reinforcing the link between increased foreign equity holdings and improved risk sharing. This time-series evidence suggests that rising international asset holdings and enhanced risk sharing are interconnected phenomena. Figure 2 and 3 estimates the year-to-year income and consumption smoothing measures, the risk sharing percentage is measured as $100 \times (1 - \beta)$. The time series of these coefficients fluctuates, so they are smoothed using HP filter. Overall, the time-series evidence indicates that the rise in international asset holdings is closely linked to increased international risk sharing. The subsequent section examines if this relationship holds across different panel data dimensions.

Panel-data regressions:

The panel data regressions are of the form:

$$\Delta \log (\text{GNI}_{it}) - \Delta \log (\text{GNI}_t) = \text{constant} + \kappa (\Delta \log (\text{GDP}_{it}) - \Delta \log (\text{GDP}_t)) + \epsilon_{it}$$

$$\Delta \log (\text{CONS}_{it}) - \Delta \log (\text{CONS}_t) = \text{constant} + \eta (\Delta \log (\text{GDP}_{it}) - \Delta \log (\text{GDP}_t)) + \epsilon_{it}$$

This regression is like the previous regression except that it is now estimated as a panel pooling the years in the sample. The $(1 - \text{coefficient})$ is the average amount of income smoothing within the group during the time period considered. Méltitz and Zumer (1999) impose structure on κ such that $\kappa = \kappa_0 + \kappa_1 \gamma_i$ where γ_i is an "interaction" variable affecting the amount of smoothing that country obtains. The estimated value of $1 - \kappa_0 - \kappa_1 \gamma_i$ measures the average amount of income smoothing obtained by the country during the time in question. This method is enhanced by allowing κ to change over time, besides including an interaction variable, as follows:

$$\kappa = \kappa_0 + \kappa_1(t - \bar{t}) + \kappa_2 (EHB_{it} - EHB_t)$$

Where $\bar{t}$ is the middle year of the sample period, EHB_{it} is the period t Equity Home Bias index of country i , and EHB_t is the Equity Home Bias the (un-weighted) average across the EHB_{it} at time t . The estimated value is of $100 \times (1 - \kappa_0)$ corresponds to the average amount of income smoothing within the group during the period t , the estimated value then measures the amount of income or consumption smoothing obtained in period t with Equity Home Bias EHB_{it} . The included time trend is to guard against the downward trending home bias measure. The parameter $(100 \times -\kappa_1)$ captures the average year-by-year increase in income risk sharing. The parameter κ_2 (which will be negative) captures the extent to which higher than average Equity Home Bias in a country lowers the amount of income risk sharing obtained by the country. In fact, $(100 \times -\kappa_2)$ can be interpreted as an "exchange ratio" that translates fractions of Equity Home Bias to percentage points of idiosyncratic shocks absorbed via income smoothing. The same panel regression is performed for the debt home bias by replacing EHB_{it} with BHB_{it} as an interaction term.

The analysis is repeated using foreign asset holdings relative to GDP. When total asset portfolios are small relative to GDP, little risk sharing can be obtained even by internationally well-diversified portfolios. In such cases, foreign holdings relative to GDP may be more relevant than home bias measure for macroeconomic income and consumption smoothing. In the case of equity:

$$\kappa = \kappa_0 + \kappa_1(t - \bar{t}) + \kappa_2(E_{it} - E_t)$$

where E_{it} represents the natural logarithm of the ratio of foreign equity assets to GDP for the country, and E_t is the average of E_{it} , the same formulation is used for debt and FDI asset holdings. In the same manner the consumption smoothing analysis is performed, the interaction terms are based on the ratio of foreign equity holdings to GDP, foreign direct investment to GDP and the ratio of foreign debt holdings to GDP. All estimations are performed as two-stage estimations that allow the error variance to differ across countries. The results are from regressions which include only country-fixed effects. Including time-fixed effects changes the results very little and this is because the aggregate values of the variables have been subtracted leaving little variation to be captured by time dummies.

Panel-data regressions results:

Table 3

Risk sharing Equity and Debt Home Bias : OECD and EU 1993-2003

	Average risk sharing	Interaction terms with GDP		
		Trend	Equity Home bias	Debt Home Bias
OECD				
Income	1.6 (89.98)	0.4 (2.57)	-13.69 (2.54)	
Risk				
Sharing	-0.89 (162.7)	0.23 (0.9)		-1.84 (0.29)
Consumption	20.93 (22.63)	2.29 (3.27)	-82.6 (4.12)	
Risk				
Sharing	6.1 (33.37)	0.65 (0.79)		14.25 (0.74)
EU				
Income	-4.15 (50.32)	0.17 (0.28)	5.75 (0.35)	
Risk				
Sharing	-6.74 (42.1)	0.03 (0.0616)		27.2 (1.71)
Consumption	6.11 (19.13)	-0.87 (0.606)	-27.3 (0.84)	
Risk				
Sharing	6.03 (19.2)	-0.52 (0.36)		44.7 (1.35)

Table 3 displays results for income and consumption smoothing as a function of Equity Home Bias and Debt Home Bias for the OECD and EU sample without Iceland and Ireland and numbers in parentheses are t-values. For most of the results the coefficient of the time trend is statistically insignificant. For income smoothing, the coefficient is significant to Equity Home Bias. The point estimates are clearly also significant in economic terms: the coefficient estimate for Equity Home Bias is -13.69, which implies that a country lowering Equity Home Bias by 0.1 will increase income smoothing by about 14 percent. This is the impact of home bias on risk sharing. The 13 EU Countries included are Austria, Belgium, Denmark, Finland, France, Germany, Greece, Italy, the Netherlands, Portugal, Spain, Sweden, and the UK. The values of coefficients are reported in percentages, numbers in parentheses are t-values.

Considering Debt Home Bias, the coefficient of -1.84 with an insignificant t-statistics. For consumption smoothing, the average estimated amount of risk sharing is much higher than income smoothing, and the impact of Equity Home Bias is estimated at -82.6 with high significance level. The coefficient is interpreted to imply that declining home bias has been associated with significant increasing consumption risk sharing. For the 13 EU countries income risk sharing in the EU is not significantly different from zero, also for the consumption risk sharing among EU countries neither trend nor home bias indices.

Table 4
Risk sharing and foreign asset holdings relative to GDP: OECD 1993-2003

	Average risk sharing	Interaction terms with GDP					
		Trend	Equity	Debt	FDI	Equity + debt	All assets
Income	3.35 (78.5)	0.38 (2.83)	1.6 (3.62)				
Risk	1.68 (77.39)	0.26 (2.06)		1.9 (2.17)			
Sharing	1.46 (86.1)	0.09 (0.73)			0.83 (2.19)		
	2.7 (74.6)	0.31 (2.44)				2.35 (2.9)	
	3.31 (73.01)	0.31 (2.48)					2.46 (3.33)
Consumption	21.5 (23.2)	2.07 (2.98)	7.26 (4.8)				
Risk	15.41 (23.5)	1.47 (2.14)		6.63 (2.6)			
Sharing	14.32 (25.4)	0.89 (1.23)			3.58 (2.28)		
	18.46 (22.42)	1.66 (2.42)				8.45 (3.47)	
	20.08 (22.31)	1.71 (2.48)					9.32 (3.94)

Table 4 examines whether the ratio of foreign asset holdings to GDP predicts income and consumption risk sharing in the OECD. The results show a positive effect of higher foreign equity (relative to GDP) on income and consumption risk sharing. The estimated coefficients have different orders of magnitude than those found for home bias which reflects that the Home Bias measures have different scales than the ratio of assets to GDP. Foreign direct investment is less related to income smoothing although the coefficient is still significant. Using the sum of equity and debt or the sum of all assets results in coefficients and t-values at the same order of magnitude as found for debt income risk sharing. For consumption smoothing the largest impact on risk sharing among the assets is from equity holdings, the result that higher asset holdings lead to more risk sharing is quite robust.

Table 5
Consumption risk sharing and foreign asset holdings relative to GDP: EU 1993-2003

	Average risk sharing	Interaction terms with GDP					
		Trend	Equity	Debt	FDI	Equity + debt	All assets
Consumption	12.94 (15.6)	-1.06 (0.62)	5.28 (1.17)				
Risk	12.63 (15.6)	-0.86 (0.5)		2.79 (0.33)			
Sharing	13 (15.4)	-1.04 (0.6)			3.91 (0.72)		
	12.65 (15.5)	-0.88 (0.52)				6.02 (0.74)	
	13 (15.3)	-0.93 (0.54)					7.06 (0.89)

In Table 5 regression the sample is now of EU (14 countries) estimating the relative assets holdings to the GDP. From Table 5 consumption risk sharing results are insignificant although the coefficients are robustly positive and of a similar order magnitude as those found for the OECD sample. Also, it shows that risk sharing increases with equity holdings within the EU, but the results are all insignificant. Countries included are Austria, Belgium, Denmark, Finland, France, Germany, Greece, Ireland, Italy, the Netherlands, Portugal, Spain, Sweden, and the UK.

Concluding remarks

The results shows that a high level of foreign portfolio assets is positively and robustly related to income and consumption risk sharing, it appears that equity and FDI assets are more important than debt although the data can't clearly separate out the effect of each group of assets. Countries could diversify ownership by increasing international assets rather than by savings to expand net assets as those countries can possibly obtain more efficient risk sharing by suitable tailoring of asset holdings.

Data

Foreign equity, debt, and foreign direct investment (FDI) asset and liability data are from Lane and Milesi-Ferretti (2022) (LMF). Domestic and international debt securities outstanding are from the Bank for International Settlements (BIS) Quarterly Review. GDP, GNI, population, consumption and Stock market capitalization are from the World Development Indicators (WDI) 1993-2004.